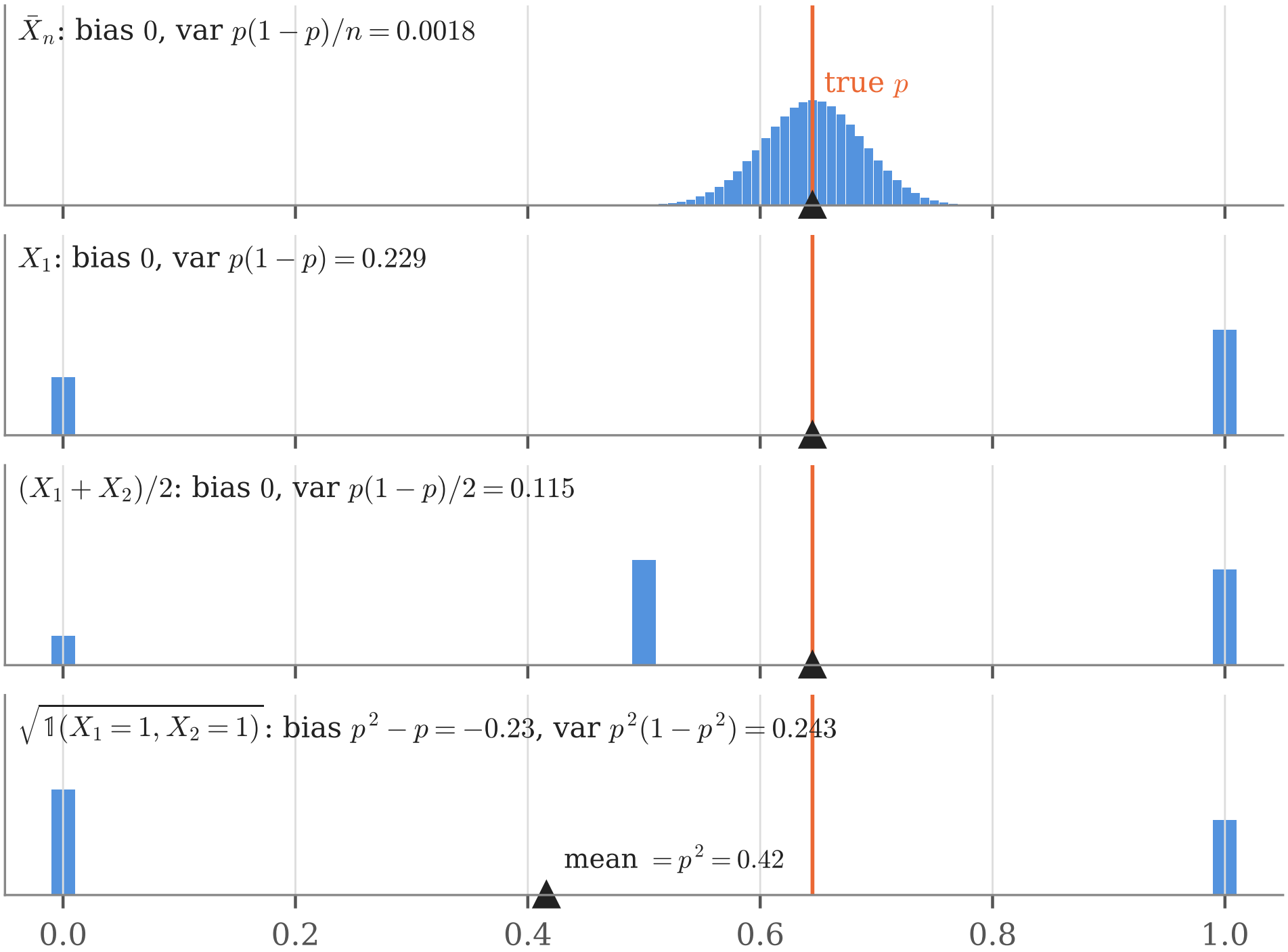
$\bar{X}_n$: bias 0, var $p(1-p)/n = 0.0018$
true p
X_1: bias 0, var $p(1-p) = 0.229$
$(X_1 + X_2)/2$: bias 0, var $p(1-p)/2 = 0.115$
$\sqrt{\mathbb{1}(X_1 = 1, X_2 = 1)}$: bias $p^2 - p = -0.23$, var $p^2(1-p^2) = 0.243$
mean $= p^2 = 0.42$
0.0
0.2
0.4
0.6
0.8
1.0
value of the estimator (orange line: the truth; triangle: the estimator's mean)